

期末考试这次是考汉译英，所以请大家注意是要你翻译汉语为英语哦！

关注这些我认为是比较重要的句子。

Unit 1

Our model provides a biologically plausible way for artificial neural networks to learn new visual concepts from a small number of examples.

我们的模型为人工神经网络从少量的例子中学习新的视觉概念提供了一种生物学上可行的方法。

The computational power of the brain's hierarchy lies in the potential to simplify learning by leveraging previously learned representations from a databank, as it were, full of concepts about objects.

大脑层级结构的计算能力在于通过利用以前从数据库中学习到的表征来简化学学习的潜力，因为数据库中充满了关于物体的概念。

Our findings not only suggest techniques that could help computers learn more quickly and efficiently, they can also lead to improved neuroscience experiments aimed at understanding how people learn so quickly, which is not yet well understood.

我们的发现不仅表明技术可以帮助计算机更快、更有效地学习的技术，还可以改进神经科学实验，这些实验的目标是了解人们是如何如此快速地学习的，因为这一点目前还没有得到很好的理解。

While there's been considerable work on inferring the goals and desires of agents, much of this work has assumed that agents act optimally to achieve their goals.

虽然已经有相当多的研究来推断智能体的目标和愿望，但大部分的研究都假设智能体会进行最佳行为以实现他们的目标。

We realized this could be used not just to speed up goal inference, but also to infer intended goals from actions that are too shortsighted to succeed, leading us to shift from scaling up algorithms to exploring ways to resolve more fundamental limitations of current AI systems.

我们意识到，这不仅可以用来加速目标推理，还可以从目光短浅而无法成功的行动中推断出预期目标，从而使我们从扩大算法规模转向探索解决当前人工智能系统更根本局限性的方法。

These complex neural hierarchies for visual recognition allow humans to learn new tasks and, crucially, leverage prior learning.

这些用于视觉识别的复杂神经层级结构使人类能够学习新的任务，并且，最为关键的是，能够利用之前的学习经验。

This is part of our larger moonshot — to reverse-engineer 18-month-old human common sense.

这是我们更大的冒险计划的一部分——对 18 个月大的人类常识进行反向工程。

This basic approach of modeling humans as imperfect reasoners feels very promising. It now allows us to infer when plans are mistaken, and perhaps it will eventually allow us to infer when people hold mistaken beliefs, assumptions, and guiding principles as well.

这种将人类建模为不完美推理者的基本方法感觉非常有前途。它现在让我们可以推断出什么时候计划是错误的，也许它最终会让我们推断出什么时候人们持有错误的信念、假设和指导原则。

Unit 2

Boston Consulting Group (BCG), a management consultancy, foresees quantum computers improving the operating income of their users by between \$450bn and \$850bn a year by 2050.

管理咨询公司波士顿咨询集团(BCG)预测，到 2050 年，量子计算机将使其用户的营业收入每年提高 4500 亿至 8500 亿美元。

Unfortunately, big, stable quantum computers do not yet exist. But small, ornery, unstable ones do.

不幸的是，大型稳定的量子计算机还不存在。但是小型不稳定的量子计算机已经问世。

A report published last year by America's National Academy of Sciences reminded readers that no commercial applications are currently known to exist.

美国国家科学院去年发表的一份报告提醒读者，目前还不存在已知的商业应用。

Doug Finke, a physicist who runs the Quantum Computing Report, which focuses on the nascent industry, is aware of 87 organisations, big and small, trying to construct quantum computers. Unlike classical computing, which had settled by the 1970s on silicon transistors as the units of computation, there is, as yet, no consensus on the best way to build a quantum computer.

负责《量子计算报告》的物理学家道格·芬克知道，有 87 家大大小小的组织在试图建造量子计算机。该报告聚焦于新兴行业。不同于 20 世纪 70 年代已确定以硅晶体管作为计算单位的经典计算，到目前为止，人们还没有就建造量子计算机的最佳方式达成共识。

The first step, then, is to go looking. The industry has been cheered by Google's demonstration last year of "quantum supremacy", in which it used a NISQ machine to perform, with minutes of computing time, a calculation that would have taken thousands of years on classical hardware.

那么，第一步就是去寻找。谷歌去年演示的“量子至上”令业界欢欣鼓舞。在演示中，该公司使用 NISQ 机器在几分钟内完成了在传统硬件上需要数千年才能完成的计算。

Google's calculation was highly contrived and of little use in the real world. But it proved a point.

谷歌的计算是高度虚构的，在现实世界中几乎没有用处。但这证明了一点（“量子至上”）。

Unlike most of its **peripatetic** predecessors, though, ByteDance has built its empire by making products that **appeal** beyond China. It is China's first global software giant. It has also **courted** foreign investors.

不过，与大多数周游世界的先行者不同，字节跳动通过制造吸引中国以外的产品来建立其商业帝国。它是中国第一个全球软件巨头。此外，它还向外国投资者示好。

In statements, TikTok has said that it stores all American “user data in the United States” and that its American operations are “not influenced by any foreign government, including the Chinese government”. The firm is trying to **mollify** critics and has hired prominent security and legal experts in America to help it.

TikTok 在声明中表示，它将所有美国的“用户数据储存在美国”，其美国业务“不受任何外国政府，包括中国政府的影响”。该公司正试图安抚批评者，并在美国聘请了著名的安全和法律专家来帮助它。

That, most lawyers and technologists reckon, may help get CFIUS off its back. Listing ByteDance in Hong Kong, which is under consideration, may also foster trust. So would anointing a non-Chinese successor—though, at 37, Mr Zhang may not be ready to retire.

大多数律师和技术专家认为，这可能有助于让美国外国投资委员会（CFIUS）不再纠缠不清。字节跳动在香港上市的计划正在考虑之中，这也可能增进信任。任命一个非中国籍的继任者也会有同样的效果——尽管 37 岁的张先生可能还没有准备好退休。

Unit 3

Major carriers are in the process of rolling out ultra-high-speed mobile-phone networks that use fifth-generation (5G) technology, transmitting data up to 50 times faster than current phone networks.

主要运营商正在铺设使用第五代(5G)技术的超高速移动电话网络，数据传输速度比目前的电话网络快 50 倍。

This deluge of data flooding through our society will enable a host of new services, including connecting new devices and equipment.

这种充斥着我们社会的数据洪流将实现一系列新服务，包括连接新设备和设备。

Around the home, that could be tomato plants in the garden that can ask for water, roofs that warn of weaknesses after extreme weather, jackets that keep parents updated on a child's location, trash cans that ask to be emptied, and milk cartons that point out expiration dates. But 5G networks can zip data back and forth quickly enough to mean that much of the processing can move to the cloud.

当然，还有物联网，它将用许多“智能”物品围绕着我们。在家里的周围，可以是花园里的番茄苗，它可以要求浇水，屋顶可以警告极端天气后的弱点，夹克可以让父母了解孩子的最新位置，垃圾桶可以要求清空，牛奶盒可以指出保质期。但 5G 网络可以足够快地来回压缩数据，这意味着许多处理可以转移到云上。

Traditionally, Qualcomm has developed and manufactured chipsets and modems that

can connect to 5G networks, but now it wants to play a bigger role in the actual infrastructure behind 5G.

传统上，高通开发和制造的是可以连接到 5G 网络的芯片组和调制解调器，但现在它希望在 5G 背后的实际基础设施中发挥更大的作用。

The speed was achieved with a bunch of fancy tech on a mmWave network and using what's called carrier aggregation, which combines multiple channels to make for faster speeds.

这一速度是通过在毫米波网络上使用一系列奇特的技术，并使用所谓的载波聚合来实现的，载波聚合将多个频道结合在一起，以实现更快的速度。

Unit 4

Neuman has found a way to fight this mismatch between a robot's "mind" and body. The method, called robomorphic computing, uses a robot's physical layout and intended applications to generate a customized computer chip that minimizes the robot's response time.

纽曼已经找到了解决机器人“思想”与身体不匹配问题的方法。这种方法被称为机器人形态计算，它利用机器人的物理布局和预期应用来生成定制的计算机芯片，从而最大限度地缩短机器人的响应时间。

“For robots to be deployed into the field and safely operate in dynamic environments around humans, they need to be able to think and react very quickly,” says Plancher.

“Current algorithms cannot be run on current CPU hardware fast enough.”

“要让机器人部署到野外，并在人类周围的动态环境中安全运行，它们需要能够非常迅速地思考和反应，”Plancher 说，“当前的算法不能在当前的 CPU 硬件上足够快地运行。”

A commonly used hardware accelerator is the graphics processing unit (GPU), a chip specialized for parallel processing.

常用的硬件加速器是图形处理器(GPU)，这是一种专门用于并行处理的芯片。

The system then designs a hardware architecture specialized to run calculations only on the non-zero values in the matrices.

然后，该系统设计一个专门的硬件架构，只对矩阵中的非零值进行计算。

Ideally we can eventually fabricate a custom motion-planning chip for every robot, allowing them to quickly compute safe and efficient motions. I wouldn't be surprised if 20 years from now every robot had a handful of custom computer chips powering it, and this could be one of them.

理想的情况是，我们最终能为每个机器人制造出定制的运动规划芯片，让它们能够快速计算出安全高效的运动。如果 20 年后，每个机器人都拥有少量的定制计算机芯片，我也不会感到惊讶，而这可能就是其中之一。

Unit 5

The ability to 3D-print bottlebrush elastomers makes it possible to leverage these unique mechanical properties in applications that require careful control over the dimensions of objects ranging from biomimetic tissue to high-sensitivity electronic devices, such as touch pads, sensors and actuators.

如果我们能够 3D 打印出制作瓶刷的弹性体，就有可能在需要严格控制应用对象尺寸的场合利用弹性体这些独特的机械性能，应用对象既可以是仿生组织，也可以是高灵敏度的电子设备，如触摸板，传感器和驱动器等。

Xie's and Mukherjee's key discovery involves the self-assembly of bottlebrush polymers at the nanometer length scale, which causes a solid-to-liquid transition in response to applied pressure. This material is categorized as a yield-stress fluid, meaning it begins as a semi-soft solid that holds its shape, like butter or toothpaste, but when sufficient pressure is applied, it liquefies and can be squeezed through a syringe. Xie 与 Mukherjee 的关键发现涉及到瓶刷聚合物在纳米长度范围内的自我组合性能，这可以使固体在受到压力时转变成液体。这种材料被归类为屈服应力流体，也就是说，它一开始是一种半软的固体，就像黄油或牙膏一样，能保持其形状，但对其施加足够的压力时，它就会液化，可用注射器挤出。

"We wanted the material and the printing process to be as clean and as easy as possible, so we played a chemistry trick with solubility and self-assembly, which enabled the solvent-free process. The fact that we don't use solvent is a tremendous advantage."

“我们希望这种材料以及打印过程尽可能干净、简单，所以就利用化学手段处理了材料的可溶性和自我组合性能，这才使整个过程无需使用溶剂。事实上，我们不使用溶剂是一个巨大的优势。”

Recycling a mix of multicolored HDPE pieces creates a dark plastic good only for

making products like park benches and waste bins, in which properties like color don't matter much.

循环利用多种颜色的 HDPE 碎片创造出一种深色塑料，只适合用于制造公园长椅和垃圾箱等产品，在这些产品中，颜色等特性并不重要。

The difficulties of recycling plastic into anything manufacturers want to use is a big reason why the world is littered with so much plastic waste, says Eric Beckman, a chemical engineer at the University of Pittsburgh.

匹兹堡大学的化学工程师埃里克·贝克曼(Eric Beckman)说，将塑料回收成制造商想要使用的任何东西的困难，是世界上充斥着如此多塑料垃圾的一个重要原因。

Since, realistically, plastic is not going away any time soon, chemists who understand the ins and outs of all this pesky plastic are working to make it easier to recycle and turn into higher-quality material that's useful for more things.

实际上，塑料不会很快消失，了解所有这些讨厌的塑料来由的化学家正在努力使其更容易回收，并将其转化为更高质量的材料，用于更多的东西。

Widespread plastics like polyethylene terephthalate, or PET, and polyethylene are so cheap to make that any breakout polymer would have a tough time entering the market, Beckman says. So for now, inherently recyclable plastic is just an academic curiosity. But maybe, decades from now, plastics made to be recyclable from the get-go will help solve the world's plastic waste problem.

贝克曼说，广泛使用的塑料，如聚对苯二甲酸乙二酯和聚乙烯，制造成本非常低，任何突破性的聚合物都很难进入市场。因此，就目前而言，固有的可回收塑料只是学术上的好奇。但也许，几十年后，从一开始就可以回收的塑料将有助于解决世界上的塑料垃圾问题。

Using heat and chemicals, it is possible to disassemble polymers into monomers, separate those building blocks from dyes and other contaminants, and piece the monomers back together into good-as-new plastic.

使用加热和化学物质，可以将聚合物分解成单体，将这些构建块从染料和其他污染物中分离出来，然后将单体重新拼接在一起，形成新的塑料。

At such high heat, the chemistry is chaotic. Plastic molecules break down randomly, generating a complex mixture of compounds that can be burned as fuel but not used to

make new materials.

在如此高的温度下，化学反应是混乱的。塑料分子随机分解，产生一种复杂的化合物混合物，可以作为燃料燃烧，但不能用于制造新材料。